embedUR

Linux Assingment-5

Name: Saravanan D
Collage: VIT University

Create a Bash script 'file_analyzer.sh', to demonstrate the following concepts:

1. Recursive functions
   - Write a recursive function to search for files in a directory and its subdirectories containing a specific keyword.
2. Redirection and error handling
   - Log errors (e.g., invalid arguments, missing files) to 'errors.log' and display them in the terminal.
3. Here document and here string
   - Use a here document to display a help menu when the '--help' option is passed.
   - Search for a keyword in a specified file using a here string
4. Special parameters
   - Use parameters like '$0', '$#', '$?' and '$@' to provide meaningful feedback.
5. Regular expressions
   - Validate inputs with regular expressions (Check if the file exists and the keyword is not empty and valid)
6. Command-line arguments using getopts
   - Use 'getopts' to handle:
   '-d <directory>': Directory to search.
   '-k <keyword>': Keyword to search.
   '-f <file>': File to search directly.
   '--help': Display the help menu.

Example usage:

```
# Recursively search a directory for a keyword
./file_analyzer.sh -d logs -k error

# Search for a keyword in a file
./file_analyzer.sh -f script.sh -k TODO

# Display the help menu
./file_analyzer.sh --help
```

Command:
nano file_analyzer.sh
chmod +x file_analyzer.sh
mkdir logs
echo “this is error message” > logs/test1.txt
echo “hello world” > logs/test2.sh
./file_analyzer.sh -f logs/test1.txt -k error
./file_analyzer.sh -d logs -k error
./file_analyzer.sh –help
./file_analyzer.sh -f wrong.txt -k error
cat errors.log

code:

```
#!/bin/bash

ERROR_LOG="errors.log"

show_help() {
cat << EOF
Usage: $0 [options]

Options:
 -d <directory>   Directory to search recursively
 -f <file>        File to search directly
 -k <keyword>     Keyword to search
 --help           Display this help menu

Examples:
 $0 -d logs -k error
 $0 -f script.sh -k TODO
EOF
}

log_error() {
```

```
    echo "ERROR: $1" | tee -a "$ERROR_LOG"
}

validate_keyword() {
    if [[ ! "$1" =~ ^[a-zA-Z0-9_]+$ ]]; then
        log_error "Invalid keyword."
        exit 1
    fi
}

search_recursive() {
    local dir="$1"
    local keyword="$2"

    for item in "$dir"/*; do
        [ -e "$item" ] || continue

        if [ -d "$item" ]; then
            search_recursive "$item" "$keyword"
        elif [ -f "$item" ]; then
            if grep -q "$keyword" "$item"; then
                echo "Found in: $item"
            fi
        fi
    done
}

if [ $# -eq 0 ]; then
    log_error "No arguments supplied."
    show_help
    exit 1
```

```
fi

# ---------- HANDLE --help ----------
for arg in "$@"; do
  if [ "$arg" == "--help" ]; then
    show_help
    exit 0
  fi
done

while getopts ":d:f:k:" opt; do
  case $opt in
    d) directory="$OPTARG" ;;
    f) file="$OPTARG" ;;
    k) keyword="$OPTARG" ;;
    \?) log_error "Invalid option"; exit 1 ;;
  esac
done

validate_keyword "$keyword"

# ---------- DIRECTORY SEARCH ----------
if [ -n "$directory" ]; then
  if [ ! -d "$directory" ]; then
    log_error "Directory does not exist."
    exit 1
  fi

  echo "Searching directory: $directory"
  search_recursive "$directory" "$keyword"
  echo "Exit status: $?"
```

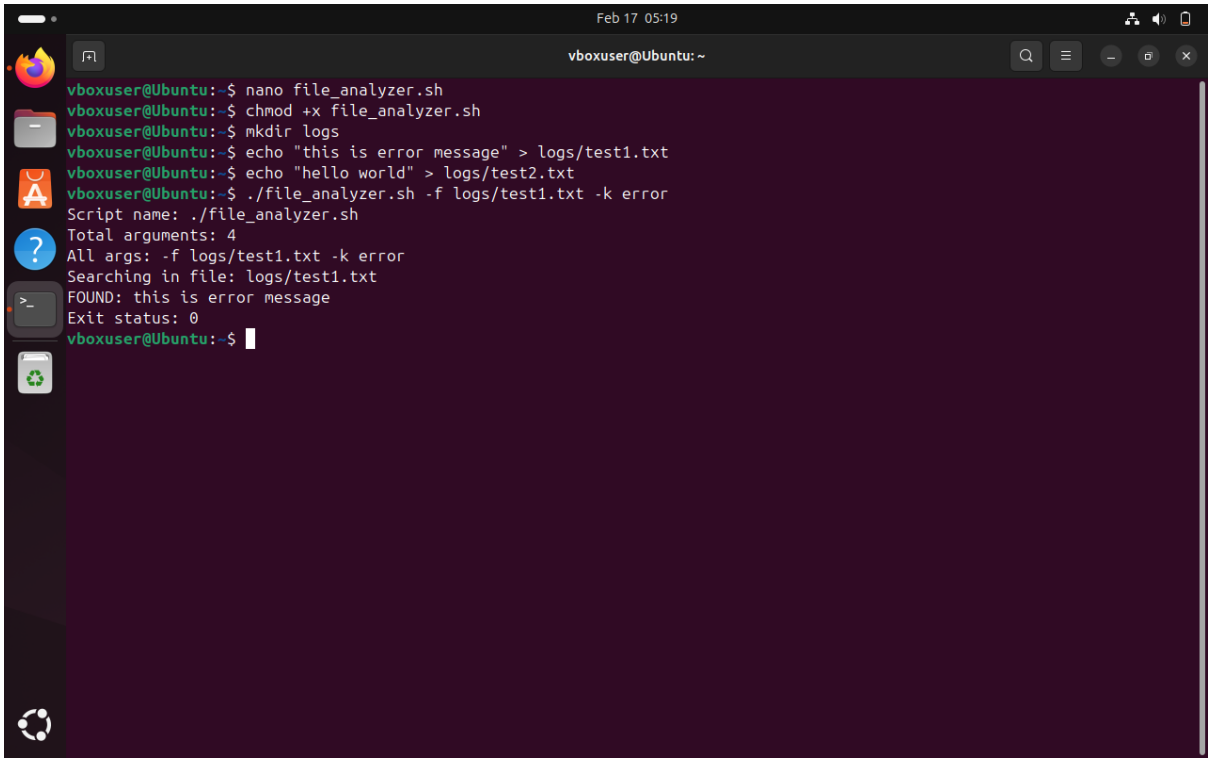
```
fi

if [ -n "$file" ]; then
  if [ ! -f "$file" ]; then
    log_error "File does not exist."
    exit 1
  fi

  echo "Searching file: $file"

  while read line; do
    grep -q "$keyword" <<< "$line" && echo "Found: $line"
  done < "$file"

  echo "Exit status: $?"
fi
```



```
vboxuser@Ubuntu:~$ ./file_analyzer.sh -f wrong.txt -k error
Script name: ./file_analyzer.sh
Total arguments: 4
All args: -f wrong.txt -k error
Tue Feb 17 05:26:03 AM UTC 2026 [ERROR] File does not exist
```

```
vboxuser@Ubuntu:~$ ./file_analyzer.sh --help
Usage: ./file_analyzer.sh [OPTIONS]

Options:
 -d <directory>   Directory to search recursively
 -k <keyword>     Keyword to search
 -f <file>        Search keyword directly in file
 --help           Display help menu

Examples:
 ./file_analyzer.sh -d logs -k error
 ./file_analyzer.sh -f script.sh -k TODO
```

```
vboxuser@Ubuntu:~$ ./file_analyzer.sh -d logs -k error
Script name: ./file_analyzer.sh
Total arguments: 4
All args: -d logs -k error
Recursive search started...
FOUND in file: logs/test1.txt
Exit status: 0
```

```
vboxuser@Ubuntu:~$ cat errors.log
Tue Feb 17 05:26:03 AM UTC 2026 [ERROR] File does not exist
vboxuser@Ubuntu:~$
```